

1. Administrative & Study Identification

Full Study Title: LuMIERE: A Phase 1/2, Multicenter, Open-label, Non-randomized Study to Investigate Safety and Tolerability, Pharmacokinetics, Dosimetry, and Preliminary Activity of 177Lu-FAP-2286 in Patients With an Advanced Solid Tumor **Short Title/Acronym:** LuMIERE **Protocol Number:** AAAT7518 **NCT Number:** NCT04939610 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** 177Lu-FAP-2286 (Therapeutic) and 68Ga-FAP-2286 (Imaging Agent) **Phase of Development:** Phase 1/Phase 2 **Medical Monitor:** Novartis Clinical Operations Lead / Investigator Dr. Fiyinfolu Balogun (Memorial Sloan Kettering), Dr. Mark Voskoboynik (Alfred Health), Dr. Ryan H. Moy (Columbia)

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** * **Phase 1:** To evaluate the safety, tolerability, and dose-limiting toxicities (DLTs) of 177Lu-FAP-2286 to determine the recommended Phase 2 dose (RP2D). * **Phase 2:** To evaluate the preliminary anti-tumor efficacy, measured by Objective Response Rate (ORR), of 177Lu-FAP-2286 as a monotherapy and in combination with chemotherapy.

Secondary Objectives: To evaluate progression-free survival (PFS), overall survival (OS), pharmacokinetics, radiation dosimetry, and to determine tumor uptake utilizing 68Ga-FAP-2286 PET/CT imaging.

2.2 Background & Rationale Fibroblast activation protein (FAP) is a cell-surface protein highly expressed on cancer-associated fibroblasts (CAFs) within the tumor microenvironment of most epithelial and mesenchymal cancers, with minimal expression in normal, healthy tissue. This distinct expression profile makes FAP a promising theranostic target for peptide-targeted radionuclide therapy (PTRT). The LuMIERE study investigates whether the targeted radiation delivery of 177Lu-FAP-2286 can effectively destroy FAP-expressing cancer cells while maintaining a manageable safety profile in patients who have exhausted alternative treatments.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm (Non-randomized) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target **\$N\$: 222 Number of Sites:** Multicenter (Global sites including US and Australia)
Estimated Study Start: 30-Jul-2021 **Estimated Primary Completion:** 30-Jun-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years old. 2. Histologically and/or cytologically confirmed advanced/metastatic solid tumor not amenable to treatment with curative intent (refractory to or progressed following prior treatments). 3. Positive tumor uptake of the imaging agent ^{68}Ga -FAP-2286 on a PET/CT scan. 4. **Phase 2 specific:** Must have cytologically/histologically confirmed advanced Pancreatic Ductal Adenocarcinoma (PDAC), Non-Small Cell Lung Cancer (NSCLC), or Breast Cancer.

4.2 Exclusion Criteria 1. Symptomatic and/or untreated central nervous system (CNS) metastases, leptomeningeal disease, or primary tumor of CNS origin. 2. Received prior radiopharmaceutical therapy or external beam radiation therapy (EBRT) to more than 25% of the bone marrow. 3. Receipt of experimental drugs, chemotherapy, or immunotherapy ≤ 14 days prior to ^{177}Lu -FAP-2286 administration. 4. Severe chronic or active HIV, HBV, or HCV infections (unless on stable antiviral therapy for > 2 weeks).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Phase 1:** Fixed IV doses ranging from 3.70 GBq to 9.25 GBq administered every 6 weeks (-1 to +7 days window). * **Phase 2 (Monotherapy/Combination):** Fixed IV dose administered at the established RP2D every 4 weeks (28 days $\pm$ 3 days).

Route of Administration: Intravenous (IV) **Duration of Treatment:** Up to a maximum of 6 doses in Phase 1; treatment in Phase 2 continues until confirmed radiographic disease progression, unacceptable toxicity, or withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Routine symptom management. In the Phase 2 combination arm, concurrent standard chemotherapy (e.g., mFOLFIRINOX for treatment-naïve PDAC patients) is permitted. **Prohibited:** Other concurrent investigational agents, angiogenesis inhibitors, or radiation therapies during the active study phase.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, RECIST v1.1 Baseline CT/MRI, 68Ga-FAP-2286 PET/CT Imaging
Baseline	Day 0	Safety Assessments, First Dose of 177Lu-FAP-2286
Interim	Every 6 Weeks (Ph 1) / 4 Weeks (Ph 2)	IV Administration, Pharmacokinetics, Dosimetry, RECIST v1.1 Efficacy Scans, Safety Labs
End of Study (EOT)	Week 6-8 post-last dose (Ph 1) / Day 28 post-last dose (Ph 2)	Final Vital Signs, Exit Interview, RECIST Evaluation, AE follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Phase 1 Definition:** Incidence of Dose-Limiting Toxicities (DLTs) to establish the RP2D. * **Phase 2 Definition:** Objective Response Rate (ORR) defined as the proportion of patients achieving a Confirmed Complete Response (CR) or Partial Response (PR). **Measurement Method:** Clinical safety labs, physical examinations, and Central/Investigator radiological assessment using RECIST v1.1.

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS) and Overall Survival (OS). * 177Lu-FAP-2286 Pharmacokinetics and Radiation Dosimetry. * Tumor FAP expression/uptake correlations via 68Ga-FAP-2286 PET/CT.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of Treatment-Emergent Adverse Events (TEAEs) graded by NCI-CTCAE criteria. **Serious Adverse Events (SAEs):** Standard reporting protocols directly to the sponsor and Institutional Review Boards. **Data Monitoring Committee (DMC):** Safety data is periodically reviewed by the Dose Escalation Review Committee (DCRC) during Phase 1 and a Data Monitoring Committee (DMC) during Phase 2 expansion.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who receive at least one dose of 177Lu-FAP-2286); Efficacy evaluable population. **Sample Size Justification:** The target sample size of 222 participants provides sufficient statistical power for the Phase 1 dose-escalation to evaluate safety, DLTs, and establish the RP2D, followed by the Phase 2 expansion to evaluate preliminary efficacy (ORR) as monotherapy and in combination with chemotherapy across selected solid tumor cohorts. **Handling of**

Missing Data: Standard multiple imputation or censoring rules applied for survival and radiographic analyses.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine site visits and remote data monitoring.

Audit Trail: Compliant eCRF data entry and image centralization for RECIST v1.1 and PET/CT reviews.

11. Study Identifiers & Governance

Primary ID: AAAT7518 **Registry Number:** NCT04939610 **Sponsor/Lead**

Organization: Novartis Pharmaceuticals **Study Title:** LuMIERE

Official Regulatory Title: A Phase 1/2, Multicenter, Open-label, Non-randomized Study to Investigate Safety and Tolerability, Pharmacokinetics, Dosimetry, and Preliminary Activity of ¹⁷⁷Lu-FAP-2286 in Patients With an Advanced Solid Tumor

12. Clinical Framework (Oncology Specific)

Primary Condition: Advanced / Metastatic Solid Tumors (including PDAC, NSCLC, Breast Cancer) **Diagnosis Threshold:** Pathologically confirmed unresectable advanced disease with positive FAP expression (confirmed via ⁶⁸Ga-FAP-2286 PET). **Medical**

Methodology: Peptide-Targeted Radionuclide Therapy (PTRT)

Optimization Target: Delivery of targeted ionizing radiation directly to cancer-associated fibroblasts (CAFs) within the TME.

13. Study Procedures & Methodology

Management Pathway: Screening -> Theranostic Imaging (⁶⁸Ga) -> Radiation Therapy (¹⁷⁷Lu) -> Post-treatment safety tracking.

Education Protocol (HCP): Radiation safety handling and targeted radionuclide therapy administration protocols. **Education Protocol**

(Patient): Radiation precaution education post-infusion. **Quality**

Audit Mechanism: Centralized radiological and dosimetry review committees.

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or withdrawal (plus long-term safety follow-up). **Onsite Visit Cadence:** Every 6 weeks (Phase 1) or every 4 weeks (Phase 2) for infusion and safety assessments. **Remote Monitoring Frequency:** As clinically indicated

for AE tracking. **Checklist Review Interval:** Radiographic evaluation every 6 weeks (42 days) in Phase 1.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	Dr. Fiyinfolu Balogun	_____	03-Apr-2026	Clinical Operations Lead	Novartis Clinical Operations Lead	_____	03-Apr-2026
Lead Statistician	Novartis Lead Statistician	_____	03-Apr-2026				